AML TRANSACTION AND FRAUD ANALYSIS

PROJECT REPORT

Prepared using	Microsoft Power BI
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1. Introduction

I developed this AML Transaction and Fraud Analysis Dashboard in Microsoft Power BI to bring transaction activity, transaction value by fraud status, fraud-status distribution, alert counts by alert type, and key account and fraud metrics into one view. The dashboard helps me review transaction activity, compare fraudulent and non-fraudulent transactions, identify changes in transaction volume, and examine alert distribution.

2. Objectives

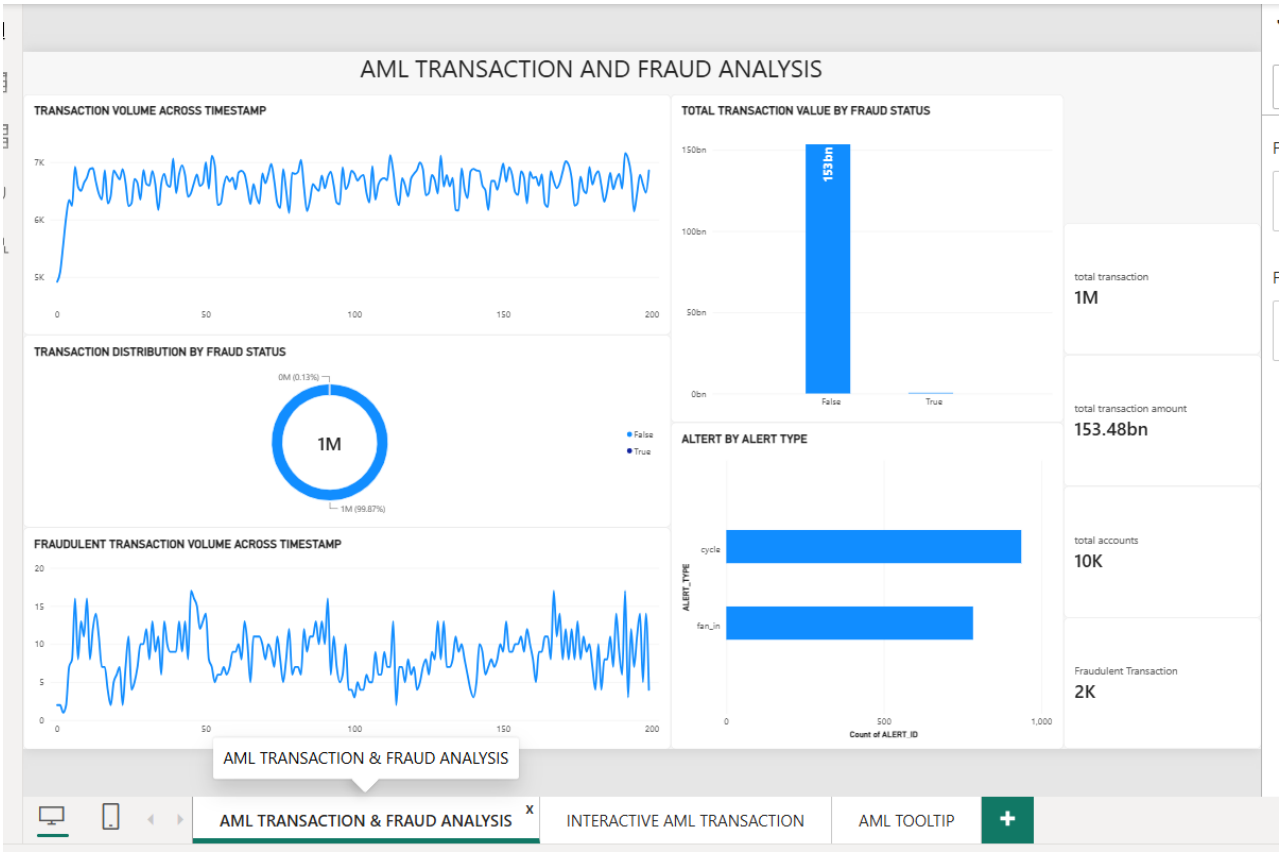
• Monitor transaction volume across the timestamp sequence.
• Compare total transaction value for fraudulent and non-fraudulent transactions.
• Analyze the distribution of transactions by fraud status.
• Compare alert counts for different alert types.
• Display total transactions, transaction amount, total accounts, and fraudulent transactions.
• Use fraud-status and alert-type filters to focus the analysis.

3. Dataset Description

I used transaction-related data containing transaction records, fraud-status information, account information, timestamps, and alert records. The dashboard includes fields and measures such as Fraud Status, Alert Type, ALERT_TYPE, ALERT_ID, transaction volume, total transaction value, and total accounts. The alert types included are cycle and fan_in.

4. Dashboard Overview

I designed the dashboard with two transaction time-series charts, a donut chart showing transaction distribution by fraud status, a column chart comparing transaction value by fraud status, a horizontal bar chart showing alert counts by type, and KPI cards. I also included filters for Fraud Status and Alert Type.



5. KPI Cards and Headline Metrics

Metric	Displayed value	Description
Total transaction	1M	Total number of transactions displayed in the dashboard.
Total transaction amount	153.48bn	Aggregate transaction amount displayed in the KPI card.
Total accounts	10K	Total account count displayed in the dashboard.
Fraudulent Transaction	2K	Number of transactions classified as fraudulent in the KPI card.

The total transaction amount is displayed as 153.48bn. The dashboard does not specify the currency or unit, so I have retained the displayed value without assigning a currency.

6. Transaction Distribution by Fraud Status

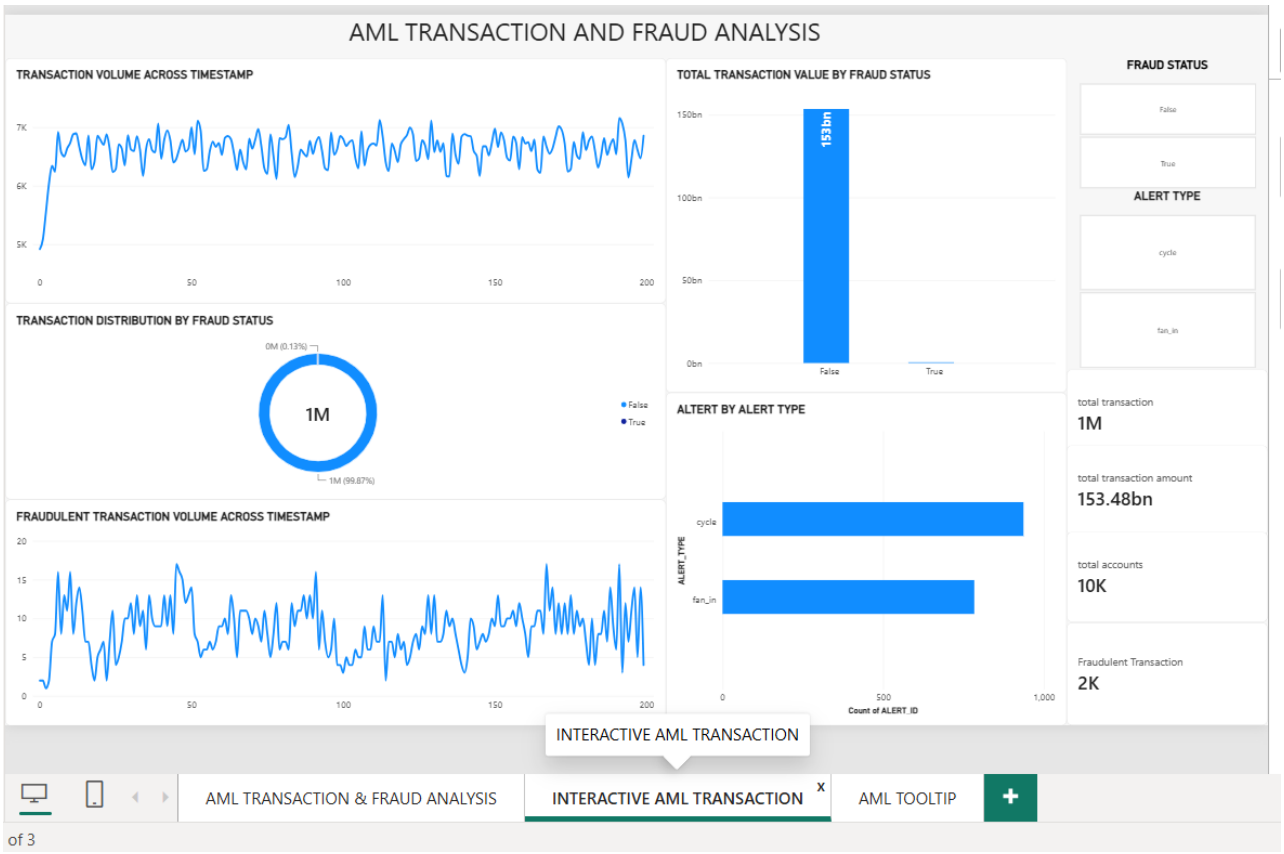
The donut chart divides transactions into False and True fraud-status categories. False accounts for approximately 1M transactions (99.87%), while True accounts for 0M (0.13%) after rounding. This shows that the non-fraudulent category represents almost the entire transaction count, while the fraudulent category represents a much smaller share. The separate Fraudulent Transaction KPI displays 2K. I have reported both values as displayed because the exact measure definitions and filter context are not available to reconcile them.

7. Transaction Volume Across Timestamp

The transaction-volume line chart shows an initial increase from approximately 5K, followed by fluctuations mainly between 6K and 7K. The series contains repeated short-term peaks and dips but remains within a relatively narrow range for most of the displayed sequence. The timestamp unit and exact dates are not identified.

8. Total Transaction Value by Fraud Status

The column chart compares total transaction value for False and True fraud-status categories. The False category accounts for almost all the displayed value, with a bar labelled approximately 153bn. The True category has a very small bar close to the baseline. The exact value for True is not displayed.



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9. Fraudulent Transaction Volume Across Timestamp

The line chart tracks fraudulent transaction volume across the timestamp sequence. The values fluctuate, generally between approximately 2 and 17, with several noticeable peaks and drops. Compared with the overall transaction-volume chart, this series has a much smaller scale and shows more irregular movement. The exact timestamp labels and aggregation interval are not identified.

10. Alert by Alert Type

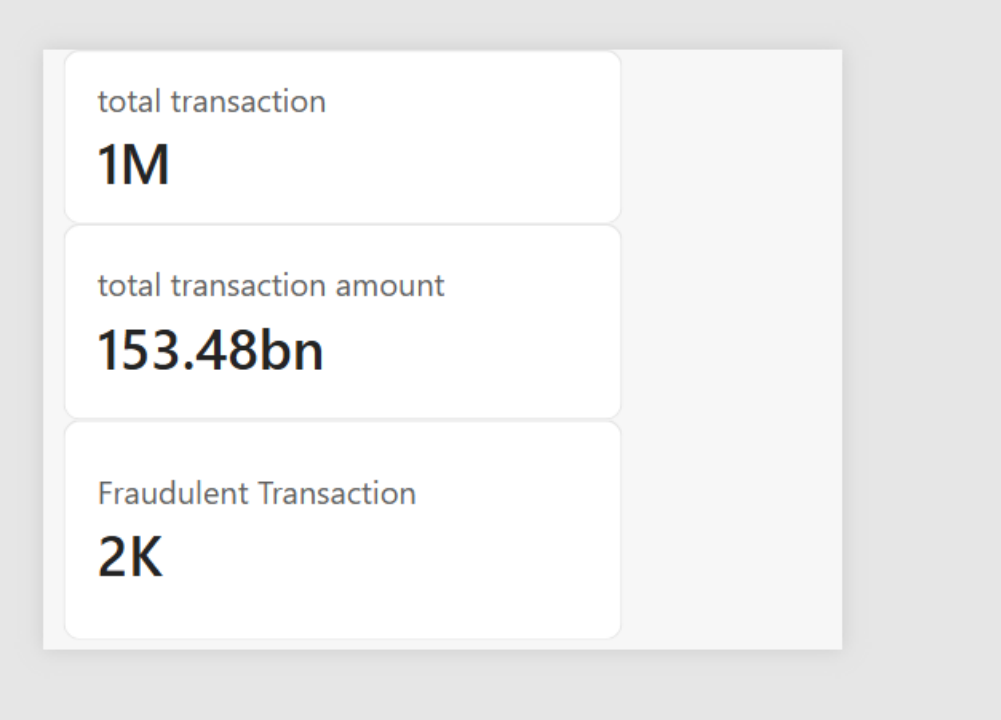
The horizontal bar chart compares the count of ALERT_ID records for cycle and fan_in. The cycle bar reaches close to 900, while fan_in reaches approximately 700. Therefore, cycle has the higher alert count in the displayed dashboard. These are approximate visual readings because exact values are not printed on the bars.

11. Filters and Interactivity

I included two filter groups in the dashboard: Fraud Status (False and True) and Alert Type (cycle and fan_in). These filters allow me to focus the dashboard on a selected fraud-status category or alert type and narrow the displayed transaction and alert information for closer analysis.

12. KPI Summary

I created a KPI summary view containing three cards: Total transaction (1M), Total transaction amount (153.48bn), and Fraudulent Transaction (2K). This view presents the main figures without the detailed charts.



13. Key Findings

• Total transactions are displayed as 1M, and total transaction amount is 153.48bn.
• The dashboard displays 10K total accounts and 2K fraudulent transactions.
• The False fraud-status category represents 99.87% of transactions, while True represents 0.13%.
• Most of the displayed transaction value belongs to the False category.
• Overall transaction volume fluctuates mainly between 6K and 7K across the plotted sequence.
• Fraudulent transaction volume varies across the sequence, generally between 2 and 17.
• The cycle alert type has a higher count than fan_in.

14. Conclusion

Through this project, I developed an AML Transaction and Fraud Analysis Dashboard in Microsoft Power BI to monitor transaction volume, compare transaction values by fraud status, analyze fraudulent transaction patterns, and review alert counts by type. The KPI cards provide a quick summary of transaction count, transaction amount, account count, and fraudulent transaction count. The analysis shows that most transactions and transaction value belong to the False fraud-status category. The True category represents a small share of the transaction distribution, while cycle is the more frequent alert type compared with fan_in. The dashboard brings these indicators together in an interactive format and supports a focused review of transaction and fraud-related activity.